

Business Insights from Exploratory Data Analysis (EDA)

1. Top-Selling Products

- The **ActiveWear Smartwatch** has the highest sales, followed closely by **SoundWave Headphones**.
- Other top-selling products include **home decor items, books, and clothing**.
- The **TechPro Textbook** has the lowest sales among the top 10 products.

2. Regional Distribution of Customers

- **South America** has the highest number of customers.
- **Europe** follows, while **North America and Asia** have slightly lower but comparable numbers.
- This suggests a strong market presence in **South America**, making it a key region for targeted marketing strategies.

3. Revenue by Category

- **Books** generate the highest revenue, indicating strong demand.
- **Electronics** follow closely, with **Clothing and Home Decor** contributing significantly.
- Despite lower revenue, **Home Decor** remains an important category for overall business profitability.

4. Signup and Purchase Trends

- Customers typically **make purchases within six months of signup**.
- This presents an opportunity to implement **early engagement campaigns** to drive conversions.

5. Customer Behavior by Region

- **North region customers exhibit recurring transactions**, making it an ideal market for **loyalty programs** or **subscription services**.
- Strategies like **personalized discounts and membership benefits** can help increase customer retention.